

Full Stack Development with MERN

Project Documentation format

1. Introduction

Project Title: ShopEZ: Online E-Commerce Application

- **Team Members:** Individual Project

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2. Project Overview

Purpose

ShopEZ is a web-based E-Commerce Shopping and Order Management System developed using the MERN Stack (MongoDB, Express.js, React.js, and Node.js). The platform enables customers to browse products, search and filter items, add products to their cart, place orders, and track purchases efficiently through an intuitive and responsive interface. The system also provides a dedicated administration panel where administrators can manage products, categories, users, orders, and promotional banners. ShopEZ aims to simplify online shopping, enhance customer experience, and provide businesses with an efficient platform for managing their online retail operations.

Features

- User Registration & Login
- JWT Authentication
- Role-Based Access Control (User & Admin)
- Product Browsing and Search
- Category-Based Product Filtering
- Product Details and Reviews
- Shopping Cart Management
- Add to Cart / Remove from Cart
- Quantity Update and Price Calculation
- Secure Checkout Process
- Order Placement and Management
- Order Tracking System
- User Profile Management
- Order History
- Admin Dashboard
- Product Management
- Category Management
- User Management
- Order Management
- Banner Management
- Sales and Revenue Analytics
- Responsive User Interface
- RESTful API Architecture
- Secure Authentication and Authorization

3. Architecture

Frontend Architecture (React.js)

The frontend of ShopEZ is developed using **React.js** following a component-based architecture. The application is designed as a Single Page Application (SPA) to provide a fast, responsive, and seamless shopping experience. React Router is used for navigation, while Context API/Redux Toolkit manages application state such as user authentication and shopping cart data.

Frontend Technologies

- React.js (Vite)
- React Router DOM
- Axios
- Bootstrap / Material UI
- Context API / Redux Toolkit
- Chart.js / Recharts

Frontend Modules

- Home Page
- User Authentication Pages
- Product Listing Page
- Product Details Page
- Shopping Cart
- Checkout Module
- Order Tracking
- User Profile
- Admin Dashboard
- Product Management
- Category Management
- Order Management
- User Management
- Banner Management

Backend Architecture (Node.js & Express.js)

The backend is developed using **Node.js** and **Express.js** and follows the **MVC (Model-View-Controller)** architecture pattern. The backend exposes RESTful APIs for authentication, product management, cart operations, order processing, and administration functionalities.

Backend Technologies

- Node.js
- Express.js
- MongoDB
- Mongoose
- JWT Authentication
- bcryptjs
- express-validator
- dotenv
- cors

Backend Modules

- Authentication Controller
- Product Controller
- Cart Controller
- Order Controller
- User Controller
- Admin Controller
- Category Controller
- Dashboard Analytics Controller

Database Architecture (MongoDB)

MongoDB is used as the NoSQL database for storing user information, product details, shopping cart data, and order records. Mongoose ODM is used to define schemas and interact with the database.

Users Collection

```
{
  "username": "string",
  "email": "string",
  "password": "string",
  "phone": "string",
  "profileImage": "string",
  "role": "USER | ADMIN",
  "createdAt": "date"
}
```

Products Collection

```
{
  "title": "string",
  "description": "string",
  "category": "string",
  "brand": "string",
  "stock": "number",
  "price": "number",
  "discount": "number",
  "ratings": "number",
  "images": ["string"],
  "sizes": ["string"],
  "featured": "boolean",
  "createdAt": "date"
}
```

Cart Collection

```
{
  "userId": "ObjectId",
  "productId": "ObjectId",
  "quantity": "number",
  "size": "string",
  "totalPrice": "number"
}
```

Orders Collection

```
{
  "userId": "ObjectId",
  "customerName": "string",
  "email": "string",
  "phone": "string",
  "address": "string",
  "paymentMethod": "string",
  "products": ["Object"],
  "totalAmount": "number",
  "orderStatus": "Pending | Confirmed | Shipped | Delivered |
Cancelled",
  "orderDate": "date"
}
```

Categories Collection

```
{
  "name": "string",
  "description": "string",
  "image": "string"
}
```

Banners Collection

```
{
  "title": "string",
  "image": "string",
  "status": "Active | Inactive"
}
```

4. Setup Instructions

Prerequisites

Before running the ShopEZ application, ensure the following software is installed on your system:

- Node.js (v18 or higher)
 - MongoDB Atlas Account or Local MongoDB Server
 - Visual Studio Code (VS Code)
 - Git
 - npm (Node Package Manager)
-

Installation Steps

Frontend Setup

```
cd client
npm install
npm run dev
```

The frontend application will run on:

`http://localhost:3000`

Backend Setup

```
cd server
npm install
npm start
```

The backend server will run on:

`http://localhost:5000`

Environment Variables (.env)

Create a `.env` file inside the **server** folder and add the following configuration:

`PORT=5000`

`MONGO_URI=mongodb://127.0.0.1:27017/shopez`

`JWT_SECRET=shopez_secret_key_987654321`

Running the Application

Open two terminals:

Terminal 1 – Backend

```
cd server
```

```
npm start
```

Terminal 2 – Frontend

```
cd client
```

```
npm run dev
```

After successful startup:

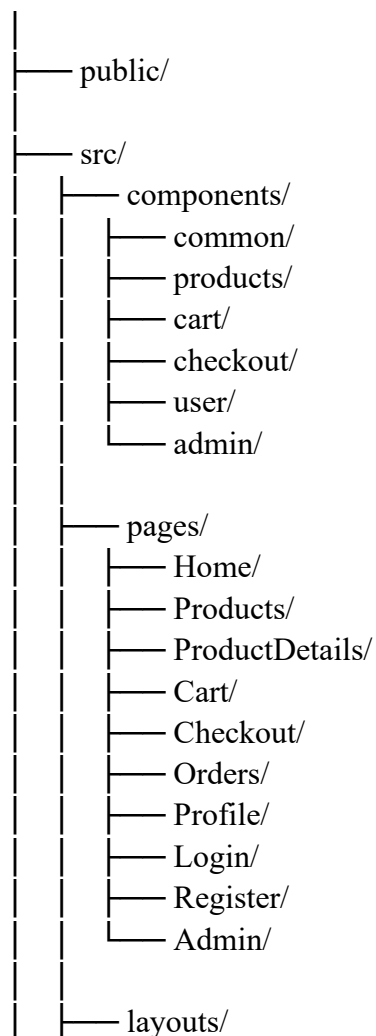
- Frontend URL: <http://localhost:5173>
- Backend URL: <http://localhost:5000>

The ShopEZ E-Commerce application is now ready for use.

5. Folder Structure

Frontend Structure (Client)

client/



```
|
|
|— services/
|— hooks/
|— context/
|— routes/
|— assets/
|— App.jsx
|— main.jsx
```

Frontend Folder Description

- **components/** – Reusable UI components such as Navbar, Product Card, Cart Item, Footer, and Admin Components.
- **pages/** – Application pages like Home, Products, Cart, Checkout, Profile, and Admin Dashboard.
- **layouts/** – Layout structures for User and Admin interfaces.
- **services/** – API service functions using Axios.
- **hooks/** – Custom React hooks.
- **context/** – Context API for Authentication and Cart State Management.
- **routes/** – Route configuration and protected routes.
- **assets/** – Images, icons, and static files.

Backend Structure (Server)

server/

```
|
|
|— config/
|— controllers/
|— middleware/
|— models/
|— routes/
|— services/
|— utils/
|— validations/
|— uploads/
|— server.js
|— .env
```

Backend Folder Description

- **config/** – Database connection and application configuration.
- **controllers/** – Business logic for authentication, products, carts, orders, and admin operations.
- **middleware/** – JWT authentication, authorization, and error handling middleware.
- **models/** – MongoDB Mongoose schemas and models.
- **routes/** – REST API route definitions.
- **services/** – Service layer for reusable business operations.
- **utils/** – Helper functions and utility methods.
- **validations/** – Request validation using Express Validator.
- **uploads/** – Product images and banner uploads.
- **server.js** – Main backend application entry point.

- **.env** – Environment variables configuration file.

6. Running the Application

To run the ShopEZ application locally, start both the frontend and backend servers in separate terminals.

Frontend

Navigate to the client directory and start the React application:

```
cd client
```

```
npm run dev
```

Frontend URL

```
http://localhost:3000
```

Backend

Navigate to the server directory and start the backend server:

```
cd server
```

```
npm start
```

Backend URL

```
http://localhost:5000
```

7. API Documentation

Authentication APIs

Register User

```
POST /api/auth/register
```

Login User

```
POST /api/auth/login
```

Product APIs

Get All Products

```
GET /api/products
```

Get Product By ID

```
GET /api/products/:id
```

Add Product (Admin)

```
POST /api/products
```

Update Product (Admin)

```
PUT /api/products/:id
```

Delete Product (Admin)

```
DELETE /api/products/:id
```

Cart APIs

Get User Cart

```
GET /api/cart
```

Add Product to Cart

```
POST /api/cart
```

Update Cart Item

```
PUT /api/cart/:id
```

Remove Cart Item

```
DELETE /api/cart/:id
```

Order APIs

Place Order

```
POST /api/orders
```

Get Orders

```
GET /api/orders
```

Update Order Status

```
PUT /api/orders/:id
```

User APIs

Get User Profile

```
GET /api/users
```

Update User Profile

```
PUT /api/users/:id
```

Admin APIs

Dashboard Analytics

GET /api/admin/dashboard

Manage Users

GET /api/admin/users

Manage Orders

GET /api/admin/orders

8. Authentication

ShopeZ implements authentication using **JWT (JSON Web Token)** to provide secure access to protected resources and user accounts.

Authentication Flow

User Login

↓

JWT Token Generated

↓

Token Stored in Local Storage

↓

Protected Routes Accessed

↓

Authorized API Requests

Security Features

- Password Hashing using bcryptjs
- JWT Token Verification
- Protected APIs
- Role-Based Authorization (User & Admin)
- Express Validator Input Validation
- Secure Environment Variables
- CORS Protection
- Rate Limiting

9. User Interface

UI Modules

User Side

- Home Page
- Login Page
- Registration Page
- Product Listing Page
- Product Details Page
- Shopping Cart
- Checkout Page
- Order Tracking Page
- User Profile Page

Admin Side

- Admin Dashboard
- Product Management
- Category Management
- User Management
- Order Management
- Banner Management
- Analytics Dashboard

UI Design Features

- Responsive Layout
- Bootstrap / Material UI Design
- Product Cards
- Search and Filter Components
- Navigation Bar
- Shopping Cart Drawer
- Dashboard Cards
- Charts and Analytics
- Toast Notifications
- Loading Skeletons

10. Testing

Testing Strategy

- API Testing using Postman
- Frontend Testing using Browser Developer Tools
- MongoDB Database Testing
- Authentication Testing
- Authorization Testing
- CRUD Operations Testing

Tested Modules

Authentication Module

- User Registration
- User Login
- JWT Verification

Product Module

- Add Product
- Update Product
- Delete Product
- Search Products

Cart Module

- Add to Cart
- Update Quantity
- Remove Product

Order Module

- Place Order
- Order Tracking
- Order Status Updates

Admin Module

- Product Management
- User Management
- Order Management
- Dashboard Analytics

Security Testing

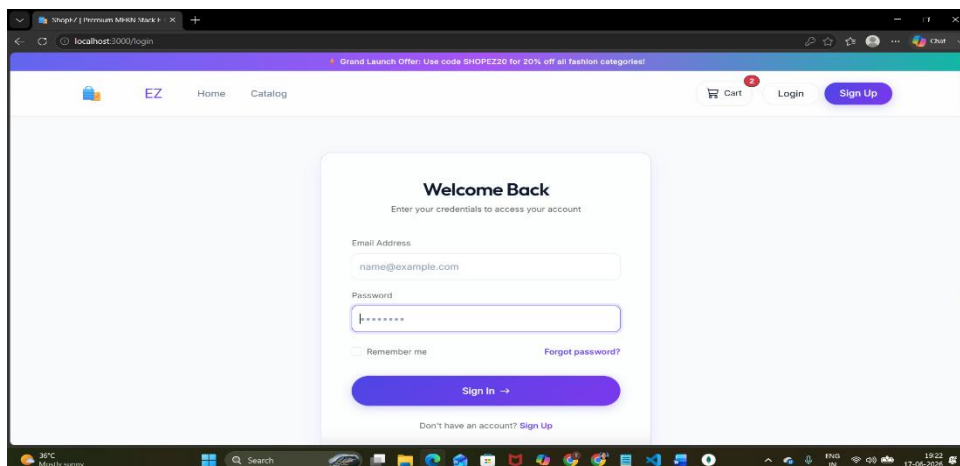
- Protected Route Access
- Unauthorized Access Prevention
- Input Validation Testing

Screenshots or Demo

Screenshots

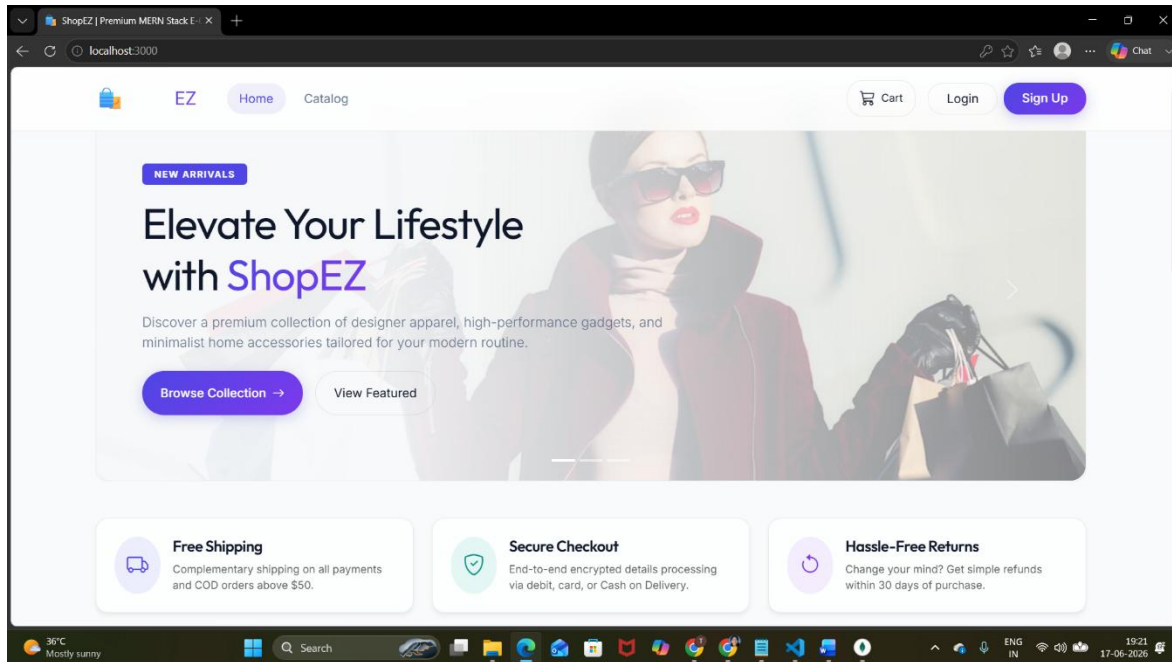
Main website screenshots

Signup page:

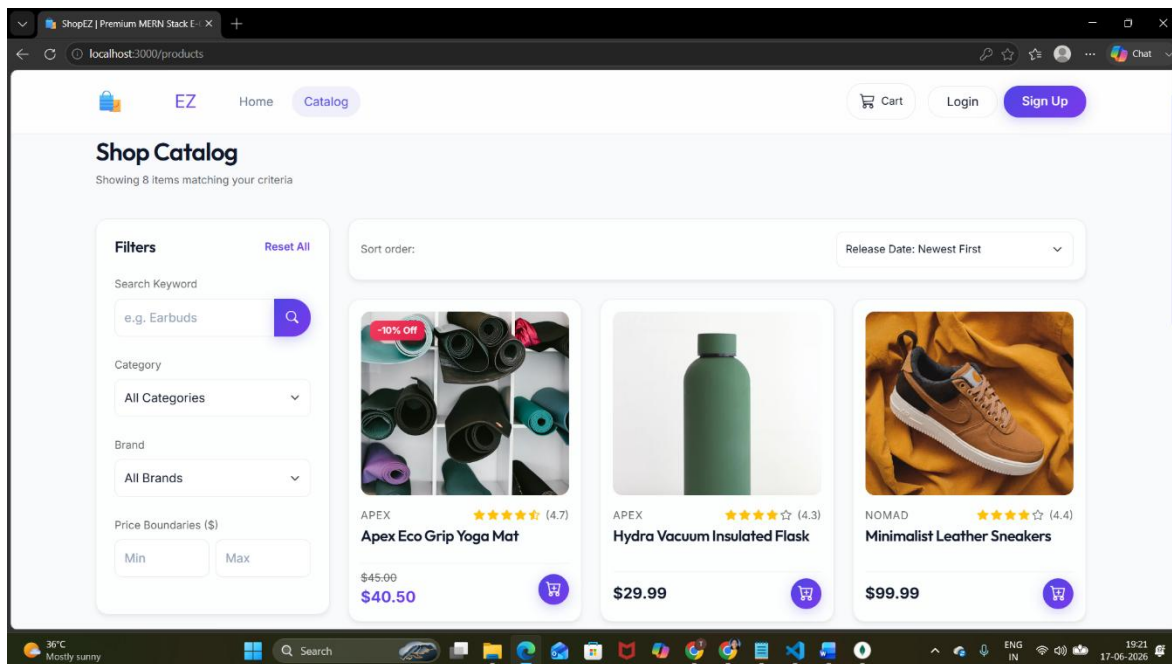


User dashboard:

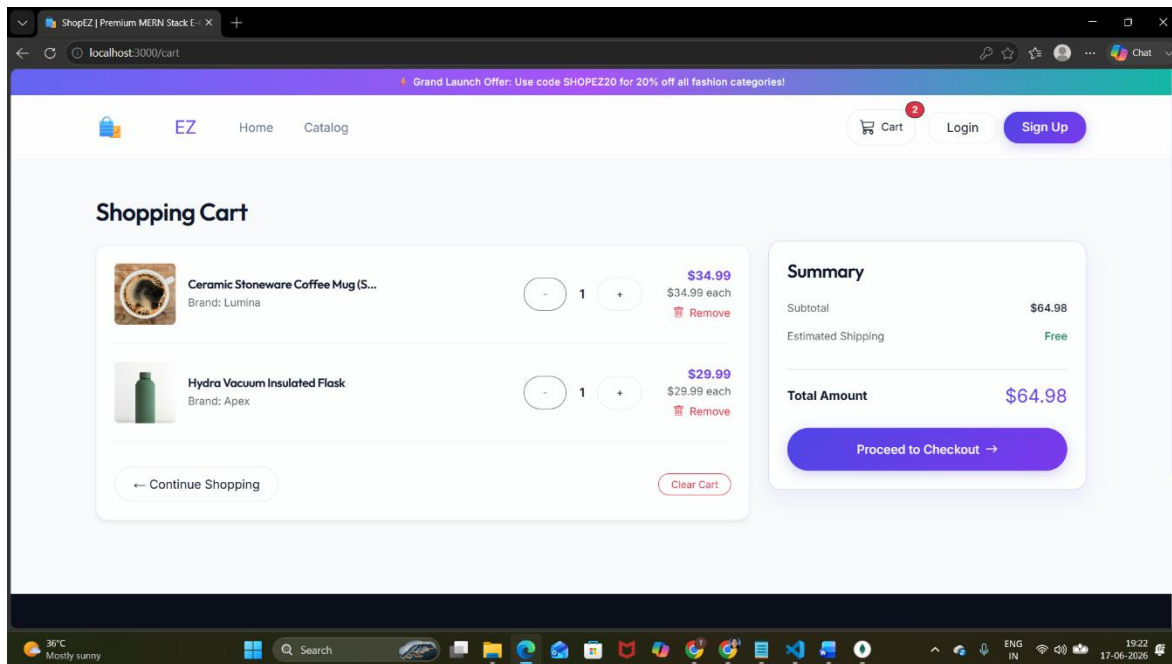
Front page:



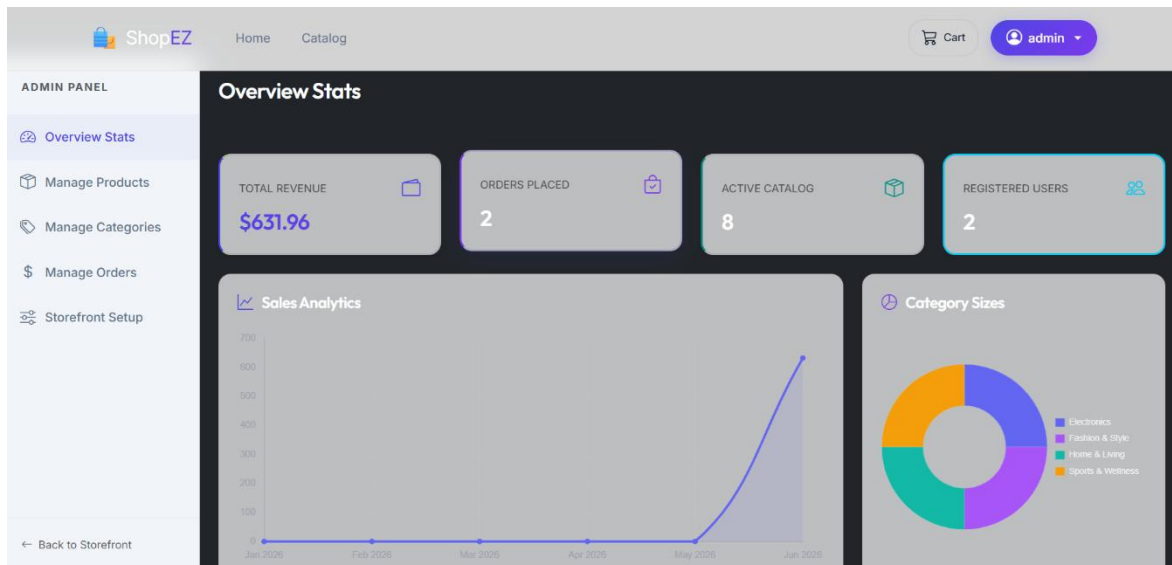
Catalog page:



Cart page:



Admin dashboard:
Status overview:



Manage product:

ShopEZ

HomeCatalog

Cart

admin

ADMIN PANEL

Overview Stats

Manage Products

Manage Categories

Manage Orders

Storefront Setup

Back to Storefront

Manage Products

+ Add Product

Image	Title	Brand	Category	Price	Disc %	Stock	Featured	Actions
	Apex Eco Grip Yoga Mat	Apex	Sports & Wellness	\$45.00	10%	35	No	
	Hydra Vacuum Insulated Flask	Apex	Sports & Wellness	\$29.99	0%	99	Yes	
	Minimalist Leather Sneakers	Nomad	Fashion & Style	\$99.99	0%	25	No	
	Lumina Arc Table Desk Lamp	Lumina	Home & Living	\$59.99	5%	40	Yes	
	Ceramic Stoneware Coffee Mug (Set of 2)	Lumina	Home & Living	\$34.99	0%	60	No	
	Nomad Premium Leather Jacket	Nomad	Fashion & Style	\$189.99	20%	14	Yes	
	Chronos Smart Watch Series 7	Chronos	Electronics	\$249.99	10%	28	Yes	
	Acoustic PRO Wireless Earbuds	Acoustic	Electronics	\$129.99	15%	50	Yes	

Manage categories:

ShopEZ

HomeCatalog

Cart

admin

ADMIN PANEL

Overview Stats

Manage Products

Manage Categories

Manage Orders

Storefront Setup

Back to Storefront

Manage Categories

+ Add Category

Image preview	Name	Slug key	Description	Actions
	Electronics	electronics	Latest premium gadgets, smartwatches, and headphones.	
	Fashion & Style	fashion	Modern clothing, designer shoes, and luxury wear.	
	Home & Living	home-living	Elegant home decor, lighting, and workspace gear.	
	Sports & Wellness	sports	High-performance fitness gear and training items.	

Manage orders:

ShopEZ

HomeCatalog

Cart

admin

ADMIN PANEL

Overview Stats

Manage Products

Manage Categories

Manage Orders

Storefront Setup

Back to Storefront

Manage Orders

Order ID	Customer Info	Order Date	Total Pay	Payment Type	Order Status	Payment Status	Actions
6a2e4833e7...	john@shopez.com	14/6/2026	\$406.97	COD	Pending	Pending	Adjust Status
6a2e3d2194...	john@shopez.com	14/6/2026	\$224.99	COD	Pending	Pending	Adjust Status

Demo

GitHub Repository Link: <https://github.com/shilpabiju1401/VIP-C2---ShopEZ-E-commerce-Application>

11. Known Issues

- Online payment gateway integration is not yet implemented.
- Product review and rating functionality is under development.
- Email notifications for order updates are not currently available.
- Advanced product recommendation features are pending implementation.
- Real-time order status updates currently rely on manual refresh mechanisms.

12. Future Enhancements

- Integration of Online Payment Gateways (Razorpay, Stripe, PayPal)
- Real-Time Order Tracking using Socket.io
- Email Notifications for Order Confirmation and Delivery Updates
- SMS Alerts for Order Status Changes
- AI-Based Product Recommendation System
- Advanced Sales and Revenue Analytics Dashboard
- Product Reviews and Ratings System
- Wishlist and Favorites Feature
- Inventory Management Automation
- Mobile Application Support (Android & iOS)
- Multi-Vendor Marketplace Functionality
- Personalized Customer Recommendations Based on Purchase History
- Coupon and Discount Management System

